Math Foundations — Notation Glossary

This is the canonical notation glossary for the math-foundations repository. Every symbol used across the **42 concept lessons** is listed below, grouped by category, with how to read it aloud, a plain-English meaning, and a link to the concept folder where it first appears. Auto-generated from `notation.json`; do not edit by hand.

Logic

Symbol	Read aloud as	Meaning	First seen in
$\forall$	for all	Universal quantifier. 'For every x in S, the property P(x) holds.' Used to make claims about every element of a set.	01-logic-and-proof
$\exists$	there exists	Existential quantifier. 'There is at least one x in S such that P(x) holds.'	01-logic-and-proof
$\exists!$	there exists a unique	Existence and uniqueness: there is exactly one element with the stated property.	01-logic-and-proof
$\neg$	not	Logical negation. ¬P is true iff P is false.	01-logic-and-proof
$\wedge$	and	Logical conjunction. P ∧ Q is true iff both P and Q are true.	01-logic-and-proof
$\vee$	or	Logical disjunction (inclusive). P ∨ Q is true iff at least one of P, Q is true.	01-logic-and-proof
$\Rightarrow$	implies	Material implication. P ⇒ Q means whenever P holds, Q holds. Vacuously true when P is false.	01-logic-and-proof
$\Leftrightarrow$	if and only if	Logical equivalence. P ⇔ Q means P and Q have the same truth value.	01-logic-and-proof
$\therefore$	therefore	Marks a conclusion that follows from previously stated premises.	01-logic-and-proof
$\because$	because	Marks a reason or premise supporting a conclusion.	01-logic-and-proof
$\vdash$	proves	Syntactic entailment. $\Gamma \vdash \varphi$ means φ is derivable from premises Γ in a formal proof system.	01-logic-and-proof
$\models$	models / entails	Semantic entailment. $\Gamma \models \varphi$ means every model of Γ is a model of φ.	01-logic-and-proof
$\square$	QED / end of proof	Marks the end of a proof.	01-logic-and-proof

Sets

Symbol	Read aloud as	Meaning	First seen in
$\in$	is an element of / in	Set membership. x ∈ S means x is an element of the set S.	00-sets-and-functions
$\notin$	is not an element of	Negated set membership.	00-sets-and-functions

Symbol	Read aloud as	Meaning	First seen in
$\subset$	is a (proper) subset of	A ⊂ B means every element of A is in B (often used for proper subsets).	00-sets-and-functions
$\subseteq$	is a subset of (or equal)	A ⊆ B means every element of A is in B; equality allowed.	00-sets-and-functions
$\supseteq$	is a superset of (or equal)	A ⊇ B is the reverse of B ⊆ A.	00-sets-and-functions
$\supset$	is a (proper) superset of	Reverse of ⊂.	00-sets-and-functions
$\cup$	union	A ∪ B is the set of elements in A or B (or both).	00-sets-and-functions
$\cap$	intersection	A ∩ B is the set of elements in both A and B.	00-sets-and-functions
$\bigcup$	union over	Indexed union over a collection of sets.	18-sigma-algebras
$\bigcap$	intersection over	Indexed intersection over a collection of sets.	18-sigma-algebras
$\emptyset$	empty set	The unique set with no elements.	00-sets-and-functions
$\setminus$	set minus / without	A \ B is the set of elements in A but not in B.	00-sets-and-functions
A^c	complement of A	The set of elements (in some ambient universe) not in A.	18-sigma-algebras
$\times$	Cartesian product	A × B is the set of ordered pairs (a, b) with a ∈ A, b ∈ B.	00-sets-and-functions
$\|S\|$	cardinality / size of	Number of elements of a (finite) set S; more generally, its cardinal.	02-counting
$\mathcal{P}(S)$	power set of	The set of all subsets of S. —𝒫(S)— = 2ˆ{—S—} for finite S.	02-counting
$\{x : P(x)\}$	the set of x such that P(x)	Set-builder notation: the set of all x satisfying property P.	00-sets-and-functions
$\binom{n}{k}$	n choose k	Number of k-element subsets of an n-element set.	02-counting

Functions

Symbol	Read aloud as	Meaning	First seen in
$f : A \to B$	f from A to B	A function f with domain A and codomain B; for every a ∈ A there is a unique f(a) ∈ B.	00-sets-and-functions
$\mapsto$	maps to	Specifies the action of a function on an element. x ↦ f(x).	00-sets-and-functions
$\circ$	composed with	Function composition: (g ∘ f)(x) = g(f(x)).	00-sets-and-functions
f^{-1}	f inverse	The inverse function (when bijective): $f^{-1}(f(x)) = x$. Also denotes preimage of a set.	00-sets-and-functions
id_A	identity on A	The function that sends every element of A to itself.	00-sets-and-functions
$\mathrm{dom}(f)$	domain of f	The set of inputs on which f is defined.	00-sets-and-functions
$\mathrm{cod}(f)$	codomain of f	The declared target set of f (not necessarily its image).	00-sets-and-functions
$\mathrm{im}(f)$	image of f	The set of actual outputs of f: {f(x) : x ∈ dom(f)}.	00-sets-and-functions

Symbol	Read aloud as	Meaning	First seen in
$f\|_S$	f restricted to S	The function f considered only on the smaller domain $S \subseteq \mathrm{dom}(f)$.	00-sets-and-functions

Number systems

Symbol	Read aloud as	Meaning	First seen in
$\mathbb{N}$	the natural numbers	Non-negative integers (0, 1, 2, ...) or positive integers, depending on convention.	00-sets-and-functions
$\mathbb{Z}$	the integers	All integers ..., -2, -1, 0, 1, 2, ...	00-sets-and-functions
$\mathbb{Q}$	the rationals	Numbers expressible as p/q with p, q integers and $q \neq 0$.	00-sets-and-functions
$\mathbb{R}$	the reals	The complete ordered field of real numbers.	00-sets-and-functions
$\mathbb{C}$	the complex numbers	Numbers a + bi with a, b $\in \mathbb{R}$ and $i^2 = -1$.	04-groups-rings-fields
$\mathbb{R}^n$	R to the n	n-dimensional real coordinate space; vectors of n real numbers.	05-vector-spaces
$\mathbb{R}_+$	the positive reals	The set of non-negative (or strictly positive) real numbers.	09-norms-and-metrics
$\mathbb{F}$	a field	An arbitrary field (typically $\mathbb{R}$ or $\mathbb{C}$); used as the scalar field of a vector space.	04-groups-rings-fields

Linear algebra

Symbol	Read aloud as	Meaning	First seen in
$\dim V$	dimension of V	Cardinality of any basis of the vector space V.	05-vector-spaces
$\mathrm{rank}(A)$	rank of A	Dimension of the image (column space) of a linear map / matrix.	06-linear-maps
$\det A$	determinant of A	Signed n-dimensional volume scaling factor of a square matrix; zero iff non-invertible.	06-linear-maps
$\mathrm{tr}(A)$	trace of A	Sum of diagonal entries of a square matrix; equals the sum of eigenvalues.	07-eigenvalues
$\langle u, v \rangle$	inner product of u and v	A bilinear (or sesquilinear) form encoding angle and length; e.g. the dot product on $\mathbb{R}^n$.	08-inner-product-spaces
$\|x\|$	norm of x	Length of a vector; satisfies positive-definiteness, scaling, and triangle inequality.	09-norms-and-metrics
$\otimes$	tensor product	Tensor product of vector spaces or vectors; bilinear factorisation of multilinear maps.	06-linear-maps
$\ker T$	kernel of T	Set of vectors mapped to 0 by linear map T; a subspace of the domain.	06-linear-maps
$\mathrm{im}\, T$	image of T	Set of outputs of a linear map; a subspace of the codomain.	06-linear-maps
$\mathrm{span}(S)$	span of S	Set of all linear combinations of vectors in S; smallest subspace containing S.	05-vector-spaces

Symbol	Read aloud as	Meaning	First seen in
T^*	T adjoint / T conjugate transpose	Adjoint of a linear map; on inner-product spaces ⟨T u, v⟩ = ⟨u, T* v⟩.	08-inner-product-spaces
$A^\top$	A transpose	Matrix obtained by swapping rows and columns of A.	06-linear-maps
λ_i	lambda i / i-th eigenvalue	Scalar λ such that T v = λ v for some non-zero eigenvector v.	07-eigenvalues
AB	A times B	Matrix product representing composition of the corresponding linear maps.	06-linear-maps
I_n	n-by-n identity	Square matrix with ones on the diagonal, zeros elsewhere.	06-linear-maps

Analysis

Symbol	Read aloud as	Meaning	First seen in
$\lim_{n\to\infty} a_n$	limit as n goes to infinity	The value a_n approaches as n grows without bound.	10-sequences-and-limits
$\sup S$	supremum of S	Least upper bound of S; the smallest number ≥ every element of S.	10-sequences-and-limits
$\inf S$	infimum of S	Greatest lower bound of S; the largest number ≤ every element of S.	10-sequences-and-limits
$\max S$	max of S	Largest element of S, when it exists.	10-sequences-and-limits
$\min S$	min of S	Smallest element of S, when it exists.	10-sequences-and-limits
$\to$	tends to / approaches	Convergence of a sequence or value: a_n → L means a_n approaches L.	10-sequences-and-limits
ε	epsilon	Small positive tolerance used in the ε-δ / ε-N definitions of limit and continuity.	10-sequences-and-limits
δ	delta	Tolerance on inputs paired with ε in the ε-δ definition of continuity.	11-continuity
∞	infinity	An idealised unbounded quantity used in limits, sums, and integrals.	10-sequences-and-limits
$\lvert x\rvert$	absolute value	Distance of a real number from 0; the canonical norm on ℝ.	10-sequences-and-limits
$d(x, y)$	distance from x to y	A metric: non-negative, symmetric, satisfying the triangle inequality.	09-norms-and-metrics

Calculus

Symbol	Read aloud as	Meaning	First seen in
$\frac{d}{dx}$	derivative with respect to x	Differentiation operator with respect to a single variable.	12-derivatives
$f'(x)$	f prime of x	Derivative of a univariate function f at x.	12-derivatives
$\frac{\partial}{\partial x}$	partial derivative with respect to x	Derivative of a multivariable function holding the other variables fixed.	13-multivariate-calculus
∇	nabla / gradient	Gradient operator: ∇f is the vector of partial derivatives of f.	14-gradient-jacobian

Symbol	Read aloud as	Meaning	First seen in
J_f	Jacobian of f	Matrix of all first-order partial derivatives of a vector-valued function.	14-gradient-jacobian
H_f	Hessian of f	Matrix of second partial derivatives of a scalar function f.	14-gradient-jacobian
$\int$	integral	Riemann (or Lebesgue) integral; signed area / total accumulation.	15-integration
$\iint$	double integral	Integral over a 2-dimensional region.	15-integration
$\sum$	sum / sigma	Indexed summation operator.	02-counting
$\prod$	product	Indexed product operator.	02-counting

Series and modes of convergence

Symbol	Read aloud as	Meaning	First seen in
$\xrightarrow{\text{a.s.}}$	converges almost surely to	Almost-sure convergence: P(X_n → X) = 1.	31-stochastic-processes
$\xrightarrow{P}$	converges in probability to	P(—X_n − X— ¿ ε) → 0 for every ε ¿ 0.	31-stochastic-processes
$\xrightarrow{d}$	converges in distribution to	Convergence of CDFs at every continuity point of the limit.	31-stochastic-processes
$\limsup_{n\to\infty} a_n$	limit superior	Largest accumulation value of a sequence; inf over n of sup of the tail.	16-series-and-convergence
$\liminf_{n\to\infty} a_n$	limit inferior	Smallest accumulation value of a sequence; sup over n of inf of the tail.	16-series-and-convergence
Cauchy	Cauchy sequence	Sequence whose terms eventually get arbitrarily close to each other; characterises convergence in complete spaces.	10-sequences-and-limits
$\sum_{n=1}^{\infty} a_n$	the series sum a_n	Limit of partial sums S_N = ∑_{n≤N} a_n; converges if the limit exists.	16-series-and-convergence
$\xrightarrow{L^2}$	converges in mean-square / Lˆ2	$\mathbb{E}$[—X_n − X—^{2}] → 0.	31-stochastic-processes

Probability

Symbol	Read aloud as	Meaning	First seen in
$P(A)$	probability of A	Probability of event A under a probability measure P.	19-probability-measure
$P(A \mid B)$	probability of A given B	Conditional probability of A given that B has occurred: P(A ∩ B)/P(B).	20-conditional-probabil
$\mathbb{E}[X]$	expectation of X	Expected value (mean) of a random variable X under its distribution.	25-expectation
$\mathbb{E}[X \mid Y]$	conditional expectation of X given Y	Best (in L^2) prediction of X given the σ-algebra generated by Y.	27-conditional-expectat
$\mathrm{Var}(X)$	variance of X	$\mathbb{E}$[(X − $\mathbb{E}$[X])^2]; spread of X around its mean.	26-variance-covariance
$\mathrm{Cov}(X, Y)$	covariance of X and Y	$\mathbb{E}$[(X − $\mathbb{E}$[X])(Y − $\mathbb{E}$[Y])]; signed measure of linear association.	26-variance-covariance
$\sim$	is distributed as	X ˜ D means X has distribution D.	23-distributions

Symbol	Read aloud as	Meaning	First seen in
i.i.d.	independent and identically distributed	Random variables that are mutually independent and share the same distribution.	21-independence
$\perp\!\!\!\perp$	is independent of	X ⫫ Y means X and Y are independent random variables / events.	21-independence
$\mathcal{F}$	script F / sigma-algebra	A σ-algebra: collection of measurable subsets closed under complement and countable union.	18-sigma-algebras
Ω	Omega / sample space	Sample space: set of all possible outcomes of an experiment.	17-sample-spaces
$\mathbf{1}_A$	indicator of A	Function that is 1 on A and 0 elsewhere; bridges sets and integrals.	25-expectation
$F_X(x)$	CDF of X	Cumulative distribution function: F_X(x) = P(X ≤ x).	23-distributions
$p_X(x)$	density of X at x	Probability density function: P(X ∈ A) = ∫_A p_X(x) dx.	24-pdf
$p(x)$	probability mass function	p(x) = P(X = x) for a discrete random variable X.	23-distributions

Distributions

Symbol	Read aloud as	Meaning	First seen in
$\mathrm{Bern}(p)$	Bernoulli of p	Distribution of a 0/1 random variable with P(X=1) = p.	23-distributions
$\mathrm{Bin}(n, p)$	Binomial of n, p	Number of successes in n independent Bernoulli(p) trials.	23-distributions
$\mathrm{Poi}(\lambda)$	Poisson of lambda	Counts of rare events with rate λ; P(X=k) = e^{-λ} λ^k / k!.	23-distributions
$U(a, b)$	Uniform on [a, b]	Continuous distribution with constant density 1/(b-a) on [a, b].	24-pdf
$\mathrm{Exp}(\lambda)$	Exponential of lambda	Memoryless waiting-time distribution with rate λ; density λ e^{-λ x}, x ≥ 0.	24-pdf
$\mathcal{N}(\mu, \sigma^2)$	normal of mu, sigma squared	Gaussian distribution with mean μ and variance σ^2.	24-pdf
$\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	multivariate normal of mu, Sigma	Joint Gaussian on $\mathbb{R}^n$ with mean vector μ and covariance matrix Σ.	26-variance-covariance
$\mathrm{Cat}(\pi)$	categorical of pi	Discrete distribution over k classes with probability vector π.	23-distributions
$\mathrm{Beta}(\alpha, \beta)$	Beta of alpha, beta	Continuous distribution on [0,1]; conjugate prior to the Bernoulli/Binomial.	24-pdf

Information theory

Symbol	Read aloud as	Meaning	First seen in
$H(X)$	entropy of X	Shannon entropy: H(X) = -$\mathbb{E}$[log p(X)]; average uncertainty / information.	36-shannon-entropy
$H(X \mid Y)$	conditional entropy of X given Y	Average uncertainty in X after observing Y.	36-shannon-entropy

Symbol	Read aloud as	Meaning	First seen in
$I(X;Y)$	mutual information of X and Y	Reduction in uncertainty about X from observing Y; symmetric in X, Y.	39-mutual-information
$D_{\mathrm{KL}}(p \parallel q)$	KL divergence from q to p	Relative entropy: 𝔼_p[log p − log q]; non-symmetric, non-negative.	38-kl-divergence
$H(p,q)$	cross-entropy of p and q	−𝔼_p[log q]; average code length using q on samples from p.	37-cross-entropy
$I(x)$	self-information of x	Surprisal: I(x) = -log p(x); information content of a single outcome.	35-self-information
log	log	Logarithm; in information theory typically natural (ln) or base 2 depending on units.	35-self-information
ln	natural log	Logarithm base e.	35-self-information
$\log_2$	log base 2	Logarithm base 2; gives entropy in bits.	36-shannon-entropy

Stochastic calculus

Symbol	Read aloud as	Meaning
W_t	Brownian motion at time t	Standard Wiener process: continuous, W_0 = 0, independent Gaussian increments.
dW_t	differential of Brownian motion	Infinitesimal Brownian increment; the driving noise of an SDE.
dX_t	differential of X at t	Infinitesimal change in a stochastic process; the LHS of an SDE in differential form.
$[X]_t$	quadratic variation of X at t	Limit of sums of squared increments; for Brownian motion, [W]_t = t.
$\langle X\rangle_t$	predictable quadratic variation of X	Compensator: predictable process making $X^2 - \langle X\rangle$ a martingale.
$\mathcal{F}_t$	filtration at time t	Increasing family of σ-algebras encoding information available up to time t.
$\int_0^t \sigma_s\, dW_s$	Itô integral	Stochastic integral against Brownian motion, defined as an L^2 limit of left-Riemann sums.
$df(X_t) = f'(X_t)\, dX_t + \frac{1}{2} f''(X_t)\, d[X]_t$	Itô's lemma	Chain rule for Itô processes: includes a second-order quadratic-variation correction.

Optimisation

Symbol	Read aloud as	Meaning	First seen in
$\arg\min_{x\in\mathcal{X}} f(x)$	argmin of f	Set of inputs at which f attains its minimum value over the search domain.	14-gradient-jacobian
$\arg\max_{x\in\mathcal{X}} f(x)$	argmax of f	Set of inputs at which f attains its maximum value.	14-gradient-jacobian
∇L	gradient of L	Gradient of a loss / objective function; direction of steepest ascent.	14-gradient-jacobian
η	eta / learning rate	Step size in gradient-based optimisation.	14-gradient-jacobian

Symbol	Read aloud as	Meaning	First seen in
softmax	softmax	softmax(z)_i = e^{z_i} / Σ_j e^{z_j}; turns logits into a probability vector.	37-cross-entropy
$\mathcal{L}$	Lagrangian	Lagrangian for a constrained optimisation; encodes objective plus multiplier × constraints.	14-gradient-jacobian
$\leftarrow$	is updated to	Assignment / update step (as in iterative algorithms).	14-gradient-jacobian

Information geometry and optimal transport

Symbol	Read aloud as	Meaning	First seen in
$W_p(\mu, \nu)$	p-Wasserstein distance between mu and nu	Optimal-transport distance: inf over couplings (X,Y) of (𝔼[d(X,Y)^p])^{1/p}.	41-optimal-transport
$F(\theta)$	Fisher information at theta	Riemannian metric on a parametric statistical model; F(θ)_{ij} = 𝔼[(∂_i log p)(∂_j log p)].	40-information-geom
$B_{\mathrm{KL}}(p, \varepsilon)$	KL ball of radius epsilon around p	Set of distributions q with D_KL(p $\|$ q) ≤ ε; trust region in info-geometric methods.	40-information-geom
$\tilde{\nabla} L(\theta)$	natural gradient of L	Fisher-preconditioned gradient: F(θ)^{-1} ∇L(θ); steepest descent in KL geometry.	40-information-geom
$\Gamma(\mu, \nu)$	couplings of mu and nu	Set of joint distributions with marginals μ and ν; the feasible set in optimal transport.	41-optimal-transport

Meta-notation

Symbol	Read aloud as	Meaning	First seen in
$:=$	is defined as	Introduces a definition: the LHS is defined to equal the RHS.	00-sets-and-functions
$\equiv$	is equivalent / equals by definition	Identity by definition; or logical equivalence between formulas.	01-logic-and-proof
$\approx$	is approximately equal to	Approximate equality; used informally and in numerical statements.	12-derivatives
$\propto$	is proportional to	Equality up to a positive multiplicative constant; common in Bayesian posteriors.	20-conditional-probability
$O(g(n))$	big O of g	Asymptotic upper bound: f = O(g) iff —f(n)— ≤ C —g(n)— for large n.	10-sequences-and-limits
$o(g(n))$	little o of g	f = o(g) iff f(n)/g(n) → 0; strictly negligible compared to g.	12-derivatives
$\Theta(g(n))$	Theta of g	Tight asymptotic bound: f = Θ(g) iff f = O(g) and g = O(f).	10-sequences-and-limits
$\Omega(g(n))$	big Omega of g	Asymptotic lower bound: f = Ω(g) iff —f(n)— ≥ c —g(n)— for large n.	10-sequences-and-limits
$\cong$	is isomorphic to / congruent to	Structural equivalence (groups, vector spaces, ...) or geometric congruence.	04-groups-rings-fields
$\neq$	is not equal to	Negated equality.	00-sets-and-functions
$\leq$	less than or equal to	Non-strict inequality.	00-sets-and-functions

Symbol	Read aloud as	Meaning	First seen in
$\geq$	greater than or equal to	Non-strict inequality.	00-sets-and-functions